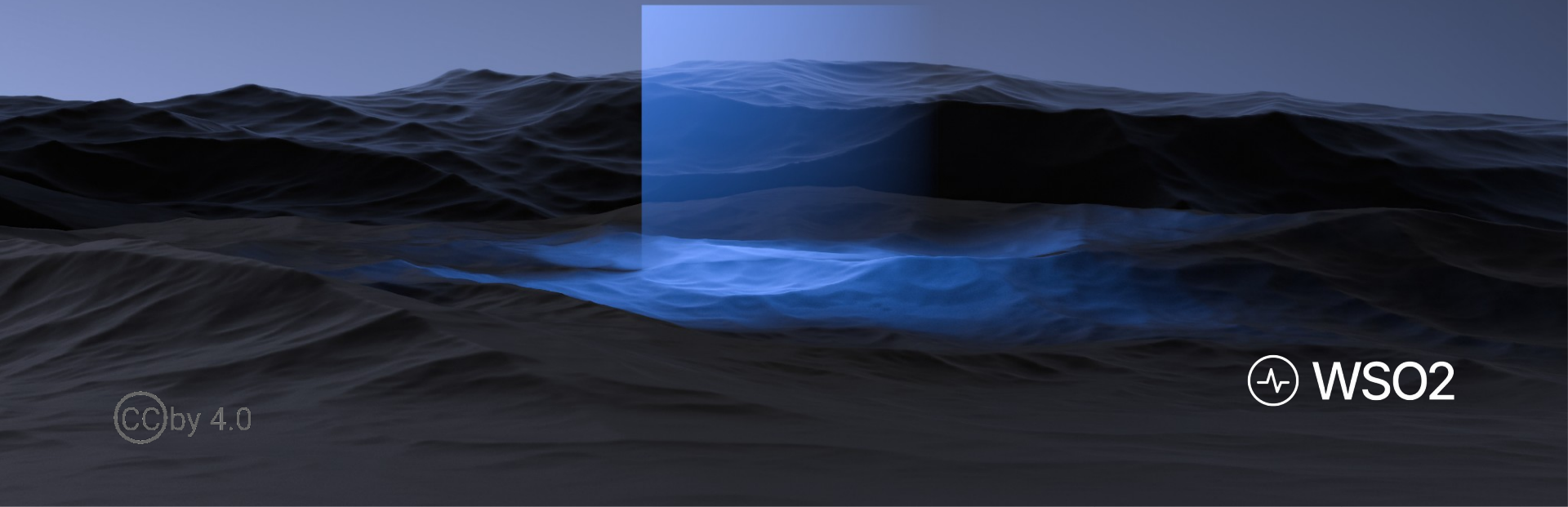
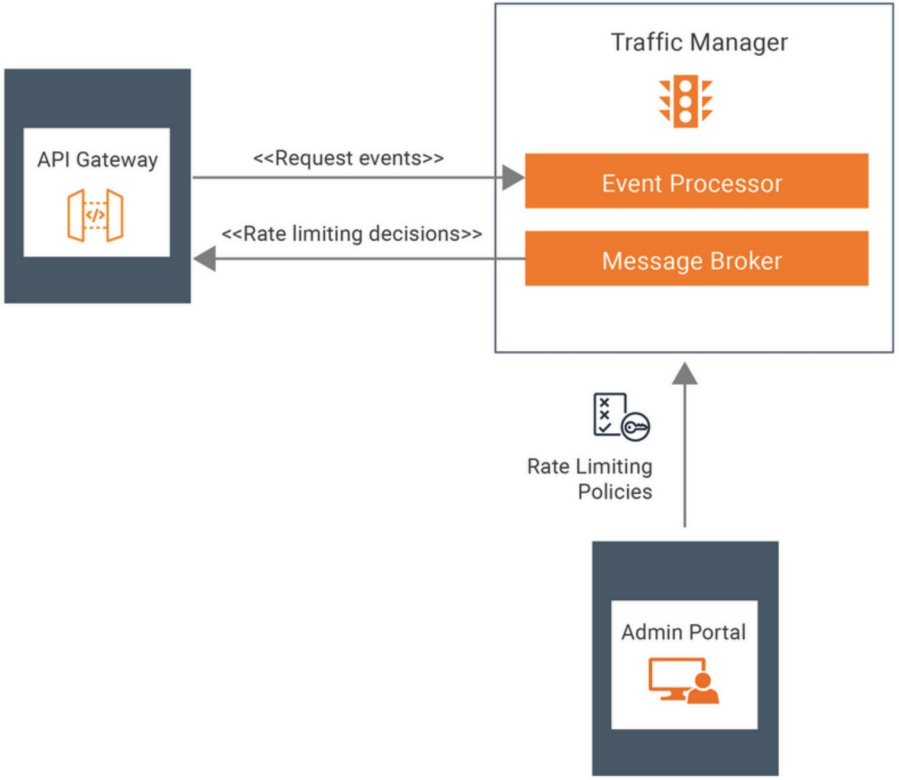
Rate Limiting



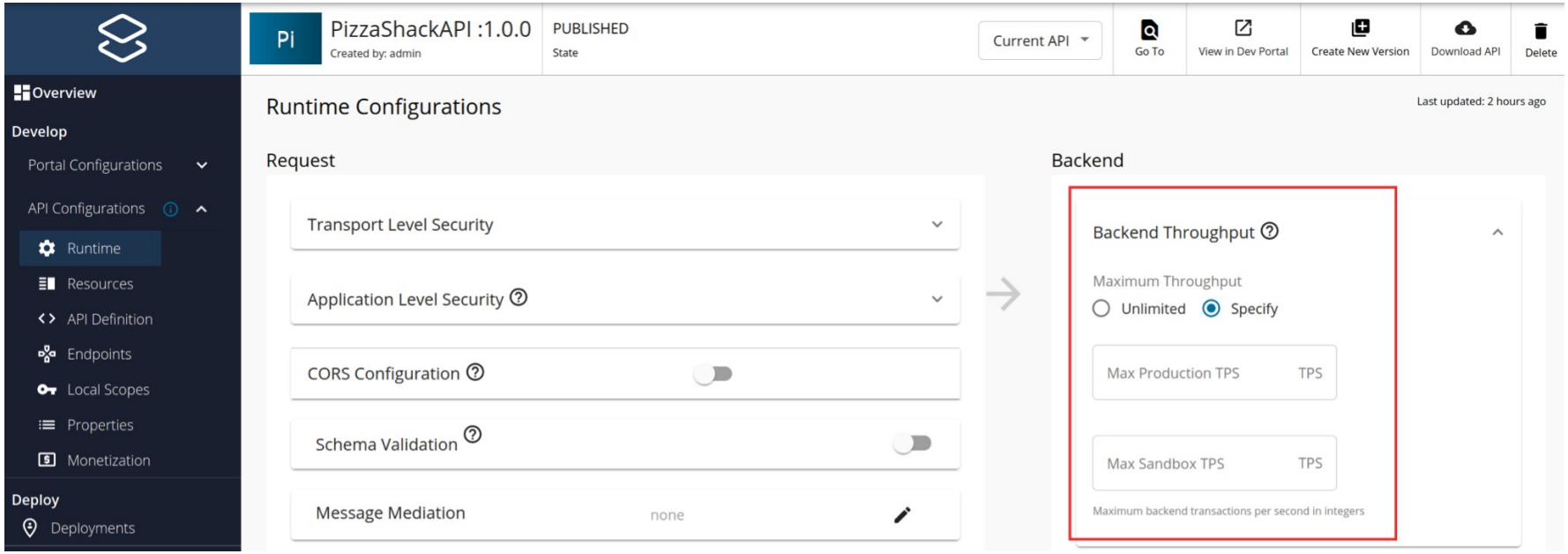
CC by 4.0

WSO2

Introducing Central Policy Server : Traffic Manager



Maximum Backend Throughput Limit



Maximum Backend Throughput Limit

Levels of Rate Limiting

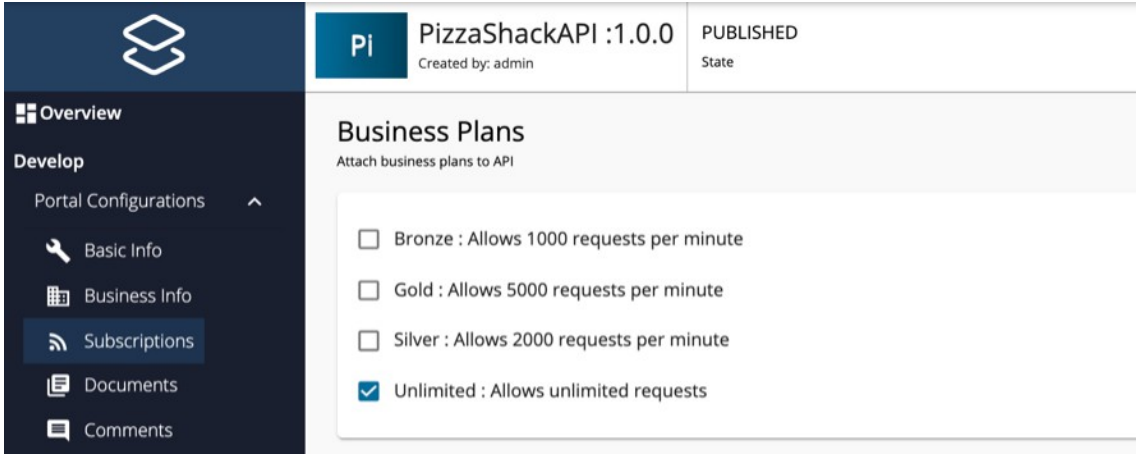
Resource Level Rate Limiting (API Publisher)	Subscription Level Rate Limiting (API Subscriber)
Application Level Rate Limiting (Application Developer)	Advanced Level Rate Limiting (API Publisher)



Subscription Rate Limiting Policies (API Publisher)

Selecting the subscription tier/s when publishing the API

- Bronze: 1000 requests per minute
- Silver: 2000 requests per minute
- Gold: 5000 requests per minute
- Unlimited: Allows unlimited access



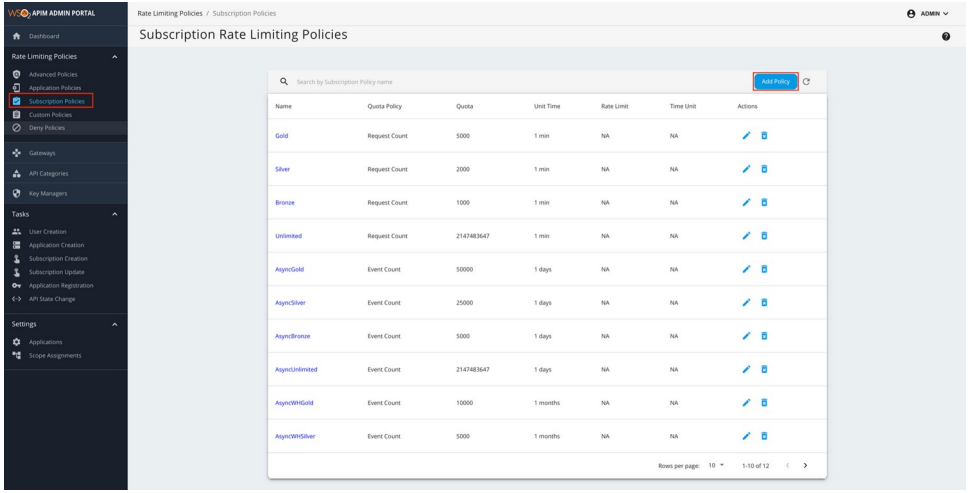
Subscription Rate Limiting Policies (API Publisher)

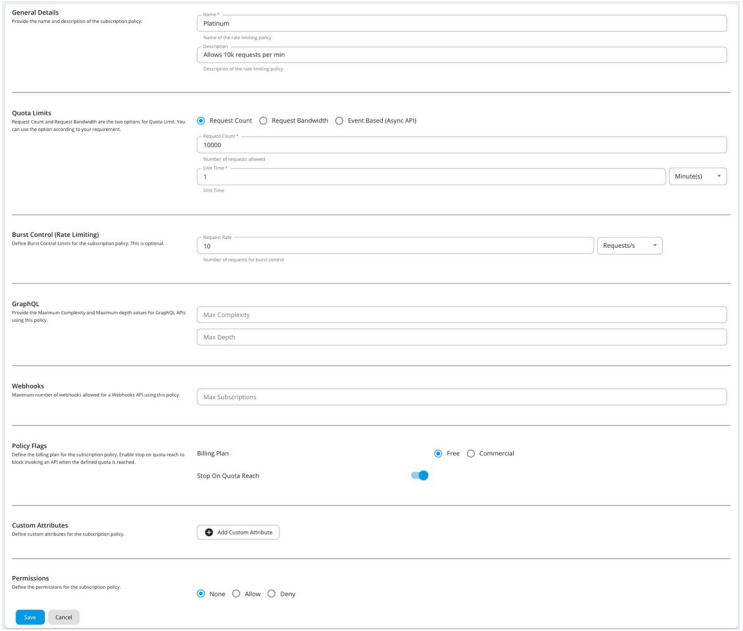
Rate Limiting (Burst Control)

- Define combined tiers
- Control the usage of APIs within smaller time durations
- Protect the backend from sudden request bursts

Adding a New Subscription Rate Limiting Policy

Admin Portal : https://localhost:9443/admin

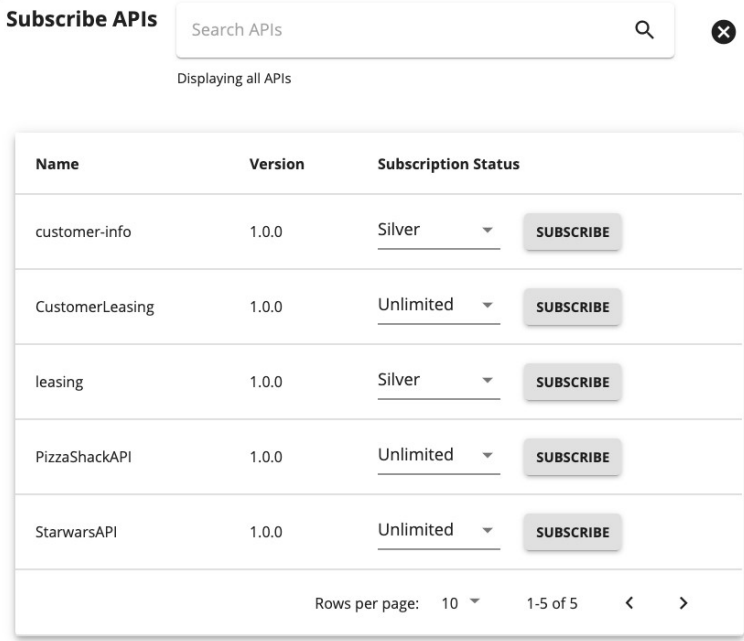




Subscription Rate Limiting Policy

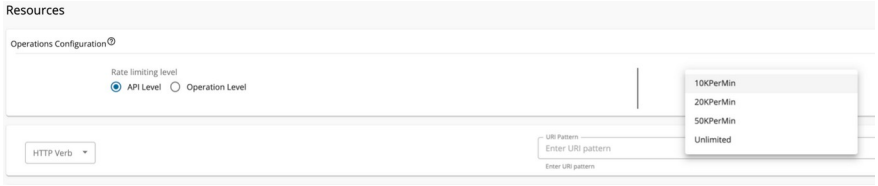
Subscription Rate Limiting (API Subscriber)

Selecting the Rate Limiting tier when subscribing to the API

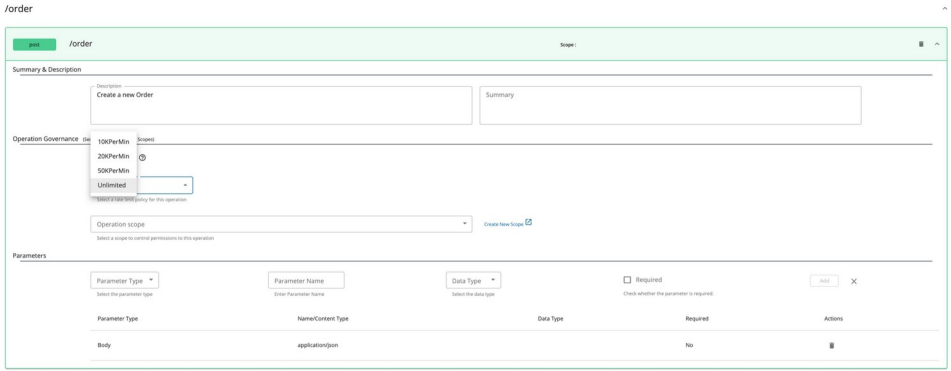


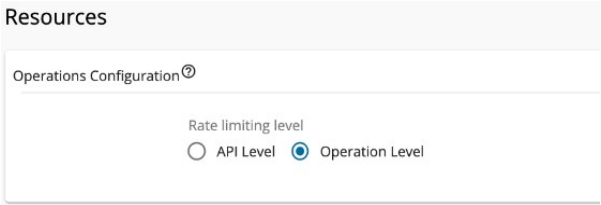
Advanced Rate Limiting (API Publisher)

Applicable in API-Level as well as Resource Level rate limiting when publishing the API



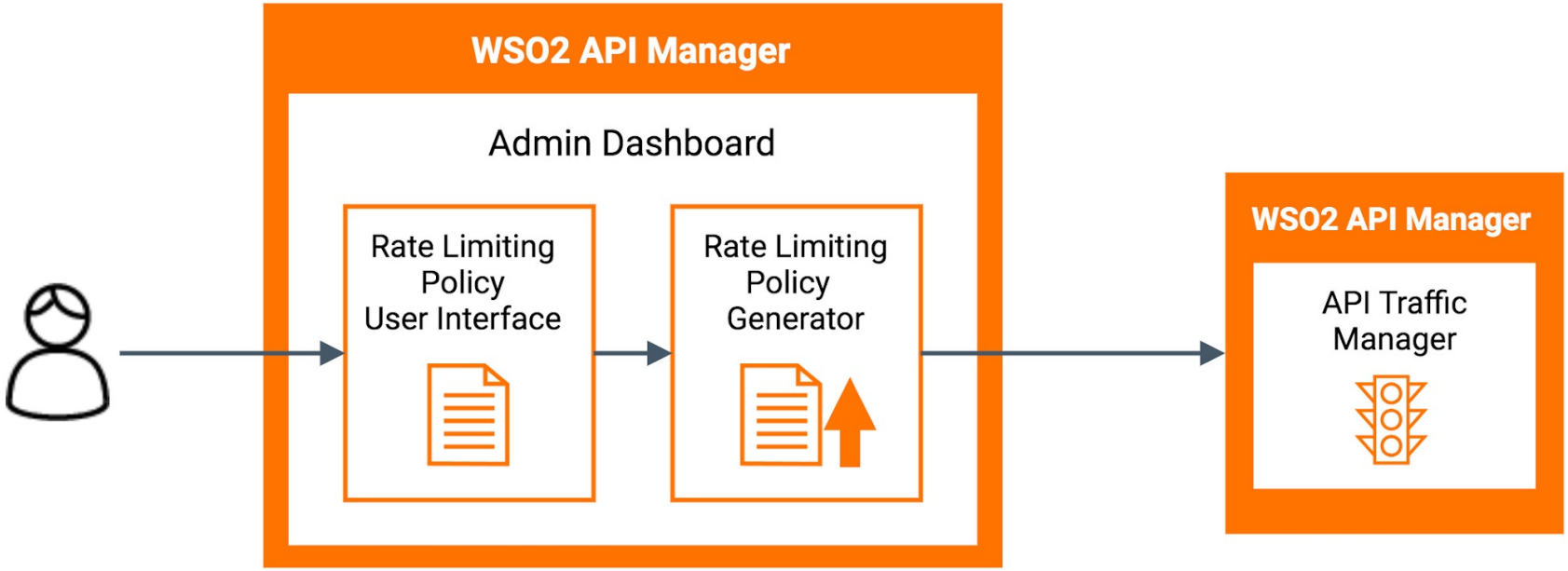
- 10KPerMin: 10,000 requests per minute
- 20KPerMin: 20,000 requests per minute
- 50KPerMin: 50,000 requests per minute
- Unlimited: Unlimited access





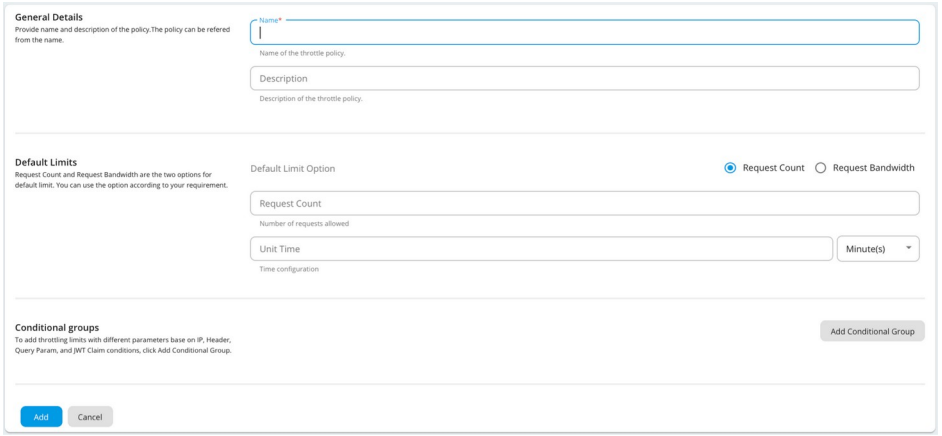
Adding New Advanced Rate Limiting Policy

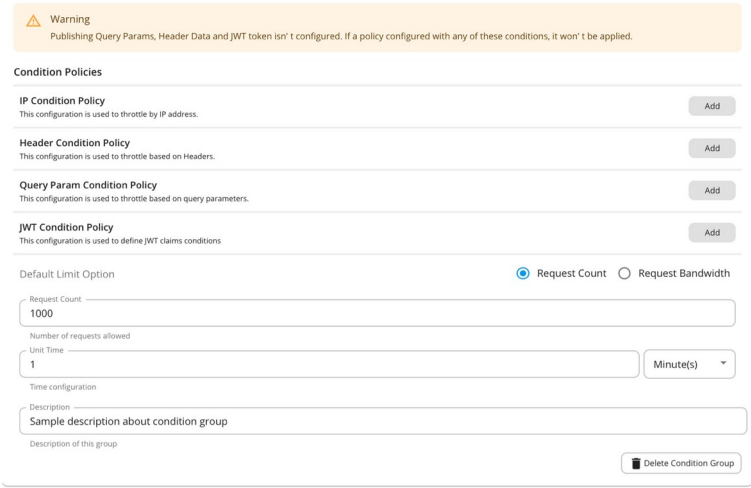
Admin Portal : https://localhost:9443/admin



Adding New Advanced Rate Limiting Policy

Admin Portal : https://localhost:9443/admin





Advanced Rate Limiting Policy

Rate Limiting Policy Definition

```
@Plan:name('carbon.super_sub_Gold')
```

Policy Name

```
@Plan:description('ExecutionPlan for sub_gold')
```

Policy Description

```
@Import('org.wso2.throttle.processed.request.stream:1.0.0')
define stream RequestStream (messageID string, appKey string, appTier string,
subscriptionKey string, apiKey string, apiTier string, subscriptionTier string,
resourceKey string, resourceTier string, userId string, apiContext string, apiVersion
string, appTenant string, apiTenant string, appId string, apiName string,
propertiesMap string);
```

Request Stream Definition

```
@Export('org.wso2.throttle.globalThrottle.stream:1.0.0')
define stream GlobalThrottleStream (throttleKey string, isThrottled bool,
expiryTimeStamp long);
```

Global Throttle Stream Definition

Rate Limiting Policy Definition

```
FROM RequestStream
SELECT messageID, ( apiTenant == 'carbon.super' and subscriptionTier == 'Gold') AS
isEligible, subscriptionKey AS throttleKey
INSERT INTO EligibilityStream;
```

Eligibility Query

```
FROM EligibilityStream[isEligible==true]#throttler:timeBatch(1 min, 0)
select throttleKey, (count(messageID) >= 5000) as isThrottled, expiryTimeStamp
group by throttleKey
INSERT ALL EVENTS into ResultStream;
```

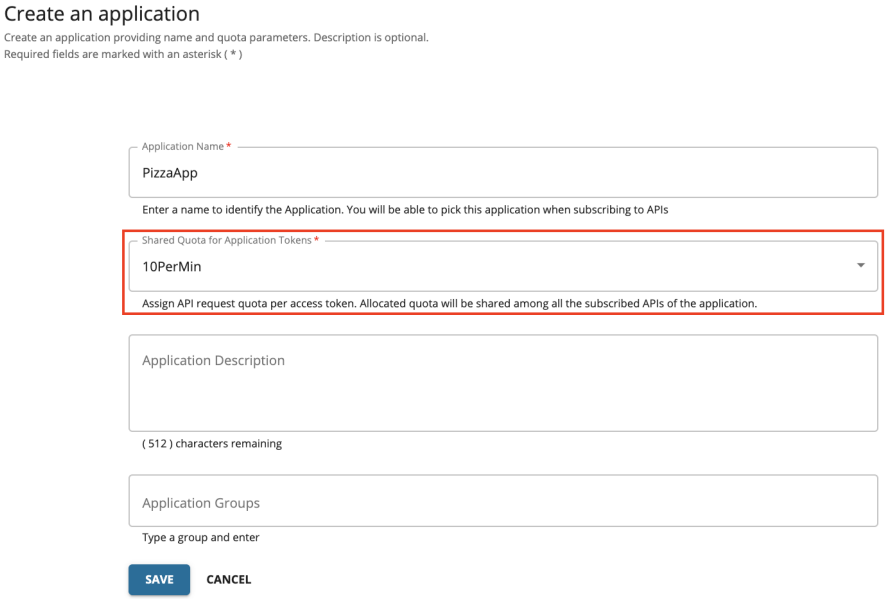
Check Rate Limit State

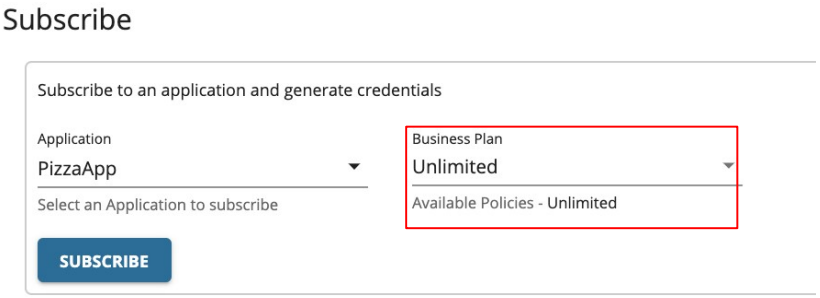
```
from ResultStream#throttler:emitOnStateChange(throttleKey, isThrottled)
select *
insert into GlobalThrottleStream;
```

Notify Rate Limiting Decision

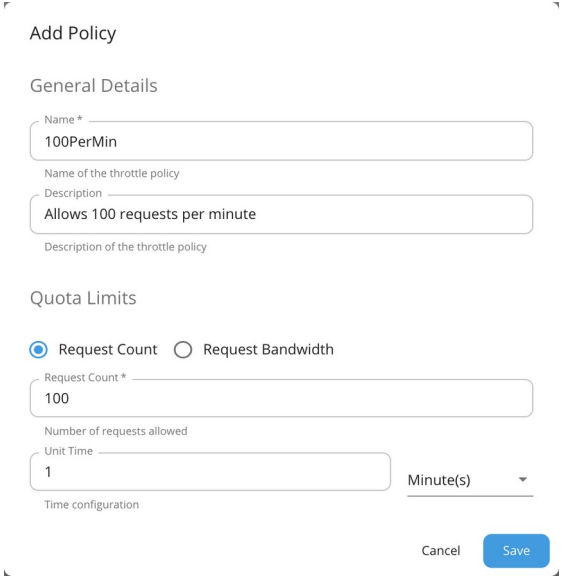
Application Rate Limiting (Application Developer)

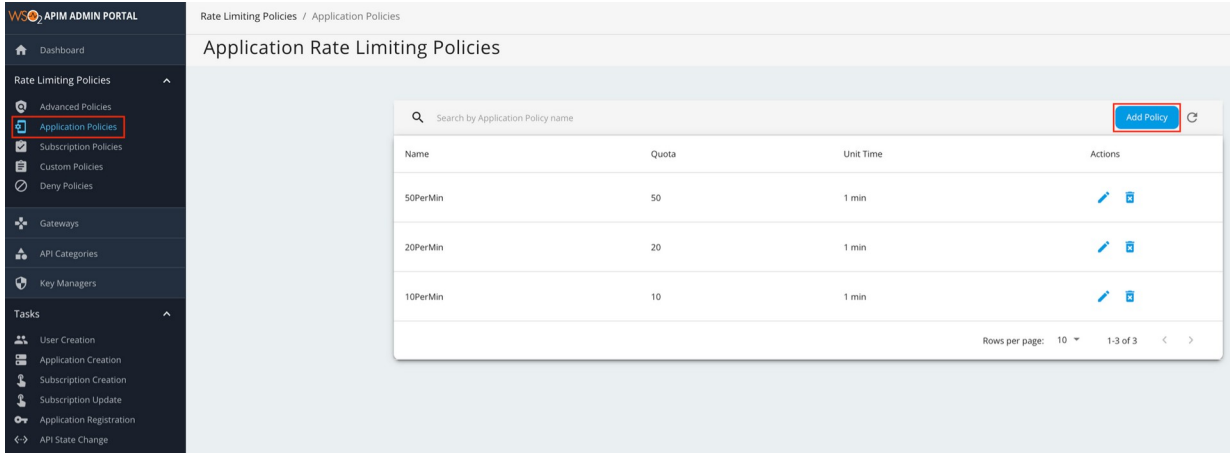
Defined when creating the application in the API Developer Portal





Adding New Application Rate Limiting Policy (Per Token Quota)

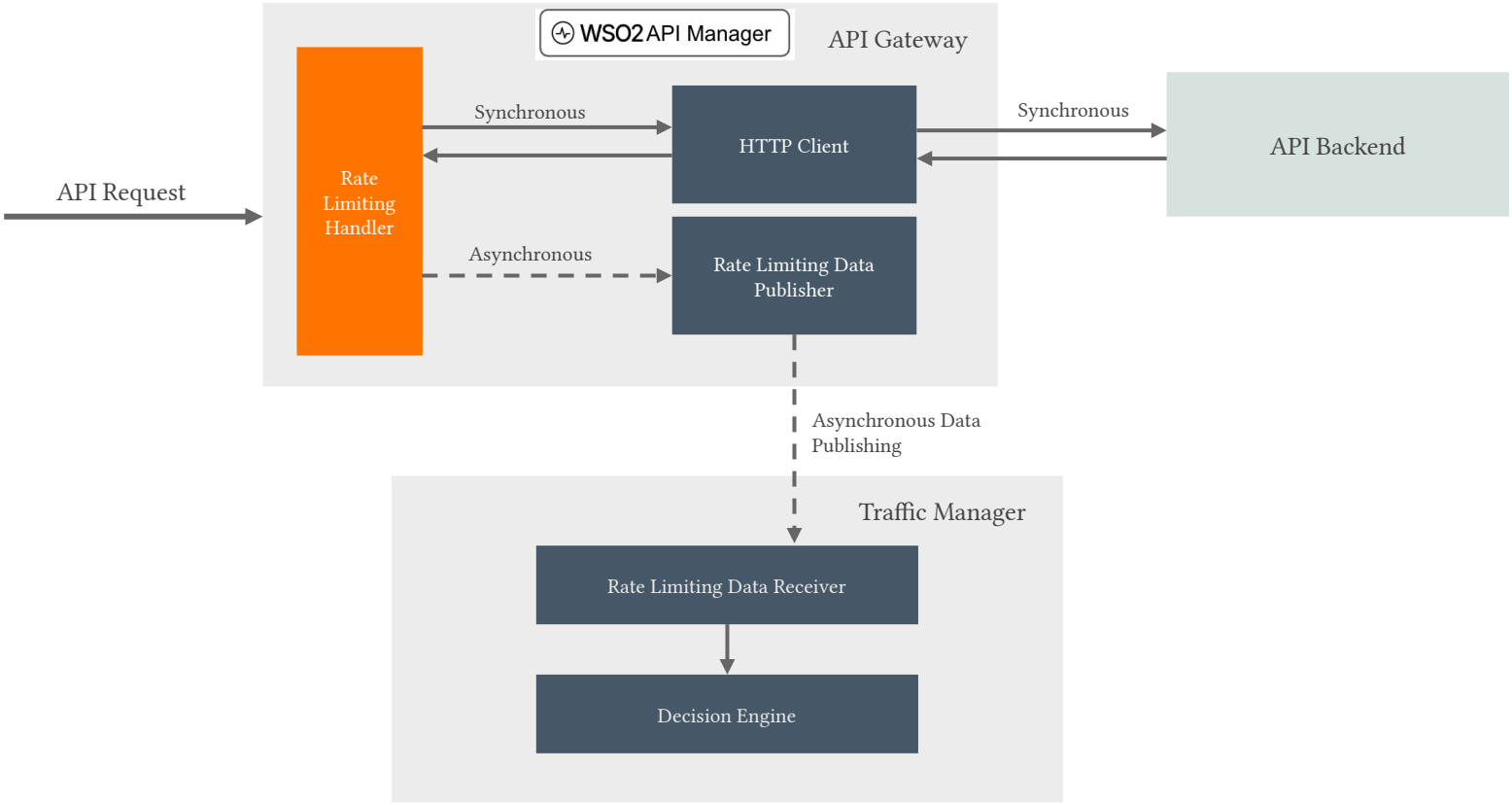




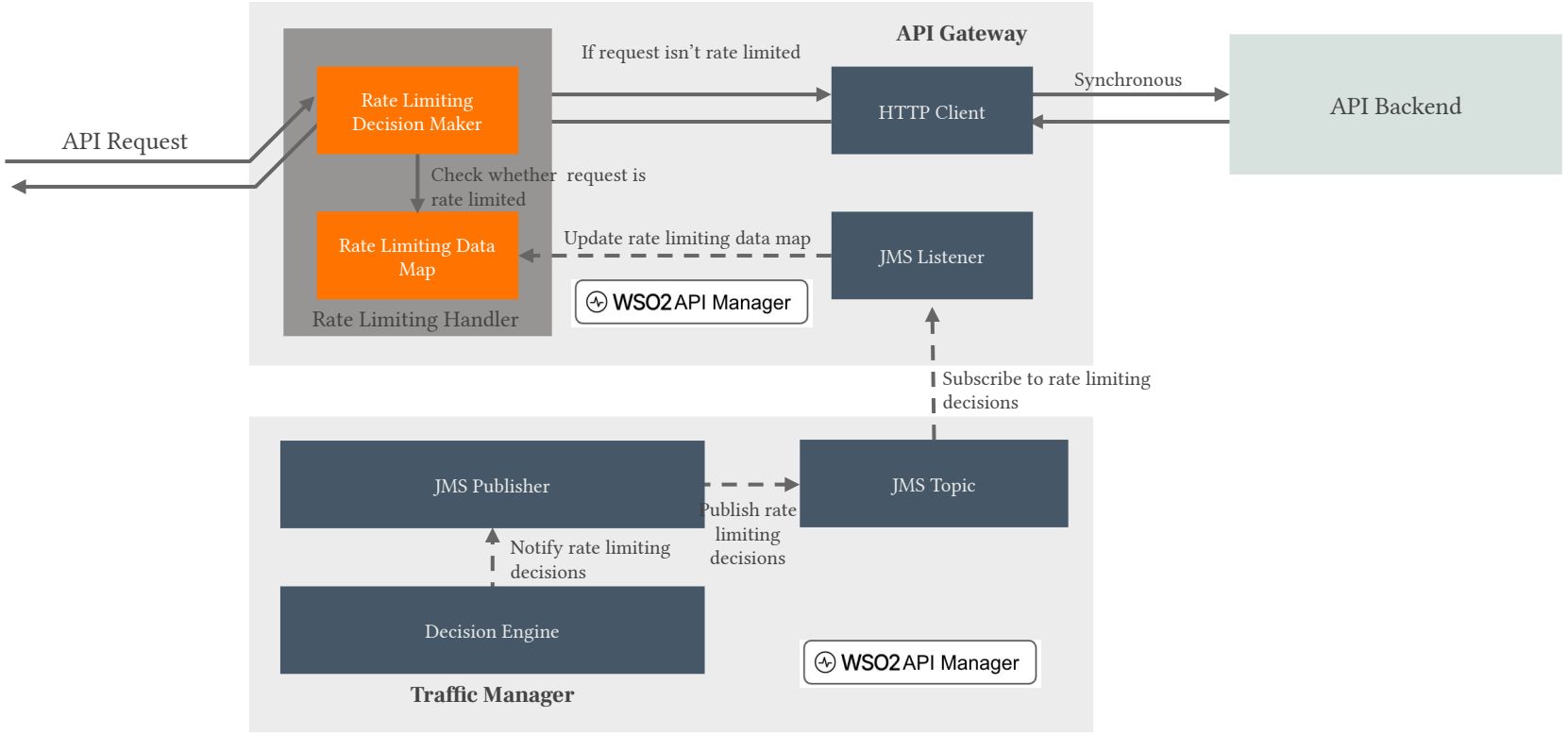
Admin Portal : https://localhost:9443/admin

Application Rate Limiting Policy

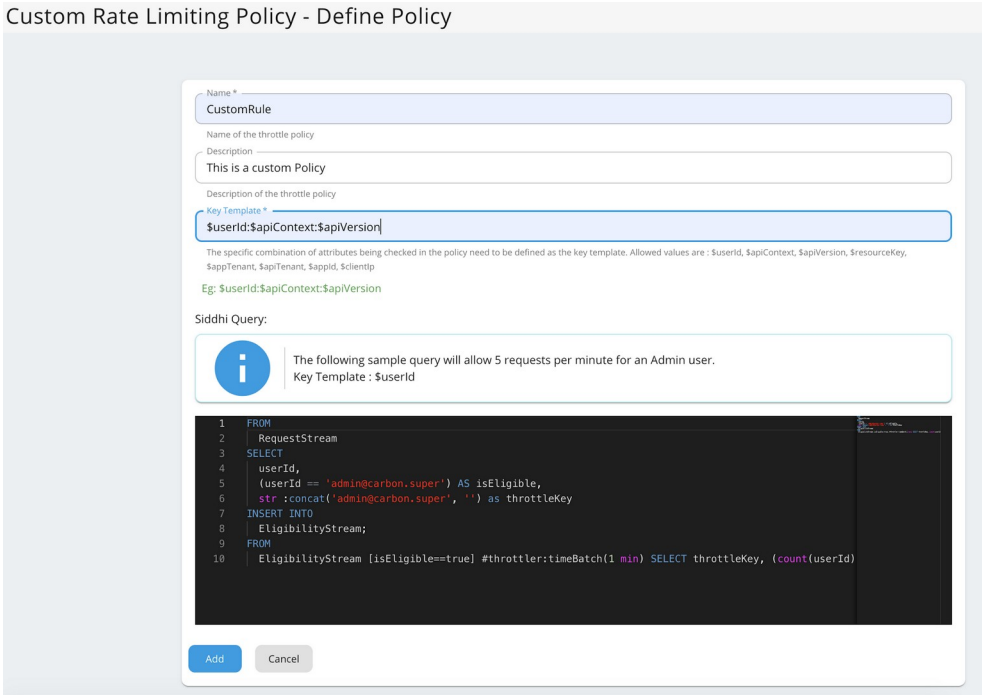
Rate Limiting Data Publishing

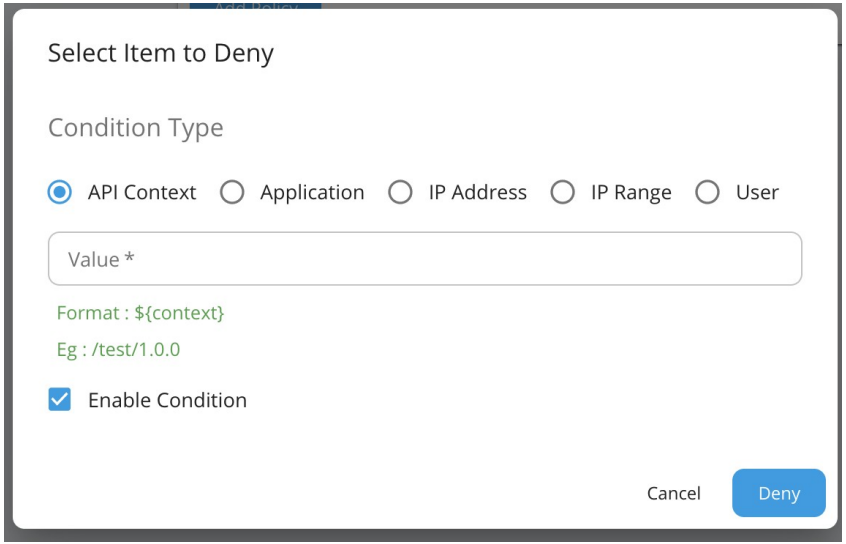


Policy Evaluation and Notifications



Custom Rate Limiting and Denying Requests





Custom Throttling

Let’s try it out!

Working with Rate Limiting Policies

WSO2